① $\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right] \xrightarrow{R2=R2+2R1} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right] \xrightarrow{R3=R3-2R2} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$x_3$ is free variable $t$, could be any #

$\boxed{\text{Yes}}$

$x_1 = 2-5t$

$x_2 = 3-4t$

② $\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R2=R2-2R1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R3=R3-R1+2R2}$

$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right] \xrightarrow[R3=-R3]{R1=R1-2R2} \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow[R2=R2+3R2]{R1=R1-9R3}$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$ $\boxed{\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}}$

③ $A(B-X) = C$

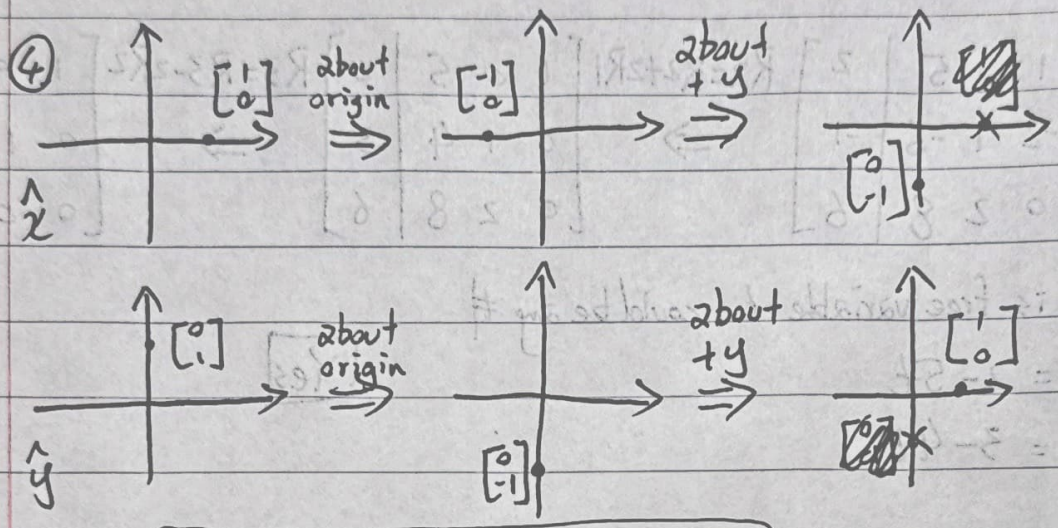
$\cancel{A(B-X) = CA^{-1}}$

$AB - AX = C$

$AX = AB - C$

$\boxed{X = A^{-1}(AB-C)}$

④




$\boxed{\text{new standard matrix is } \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}}$

⑤ $\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right] \xRightarrow[R3=R3-3R1]{R2=R2-2R1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \xRightarrow{R2=-R2+R3}$

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -4 & -10 \\ 0 & 3 & -11 & -27 \end{array}\right] \xRightarrow{R3=R3-3R2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & 1 & 3 \end{array}\right]$

$\boxed{z=3}$ ①

$y+-4z=-10$ ②

$x+y+2z=9$ ③

③-② $= x+6z=19$

$x=19-6\cdot 3$

$\boxed{x=1}$ ①

② $y=-10+4\cdot 3$

$\boxed{y=2}$

⑥ extra credit

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 1 & 0 & 2 & b_2 \\ -6 & -6 & 3 & b_3 \end{array}\right] \xrightarrow{R3=R3+3R1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 1 & 0 & 2 & b_2 \\ -3 & 0 & -6 & b_3+3b_1 \end{array}\right] \xrightarrow{R3=R3+3R2}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 1 & 0 & 2 & b_2 \\ 0 & 0 & 0 & b_3+3b_1+3b_2 \end{array}\right]$$

if $b_3+3b_1+3b_2 \neq 0$, then no solution.
but if $b_3+3b_1+3b_2 = 0$, then $Ax=b$ is true w/ one of the variable being free variable.